Homework — Week 1

AI Training

Mindset & Effective Usage

Lecture Notes and TDD Practice

October 12, 2026

Part I — Lecture Notes

The following pages reproduce the handwritten lecture notes provided for this homework.

Effective AI use starts with a clear goal, relevant context, and explicit constraints. Treat generated answers as drafts: check them against reliable sources and your own requirements instead of accepting them automatically. In coding, test-driven development turns this mindset into a repeatable cycle: write a test, observe what fails, make a focused change, and rerun the tests. Distinguishing setup errors from code failures helps make test results meaningful.

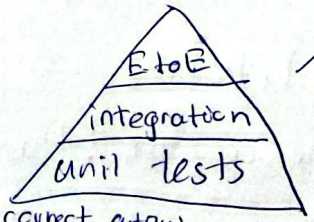
Research chat: Gemini conversation on this homework(https://share.gemini.google/pDVh28COZwN3).

TDD: Test - Driven - Dep → Test before the code exists

Red → Green → Regactor

- Red: write a test (turning red)
- Green: write the minimum amount of code to pass the test
- Regactor: Safty net → you can clear your code

The testing Levels (The paramid)



10%
20%
70%

- Unit Tests: smallest piece of code
  Example: checking if a function returns the correct output
- Integration Tests: Verify different part of your system work together
- End to End Tests: simulate a real user interacting deployed system

AI Validation:
- Anchoring Hallucinations:
- Preventing Regressions:
- Test - Driven - Prompting:

Testing:
- CLI Testing: CLI: Command Line Interfaces?
  Why? Interact directly with the OS enviroment

Detail:

- Testing Level

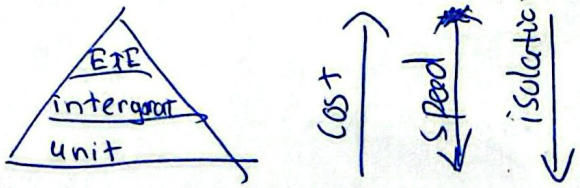
+) Unit tests: Kiểm thử mức đơn vị
- Kiểm tra xem một hàm / phương thức / UI component hoạt động như ý hay không. Example: Write unit test for shopping cart.
- How it works: Mock all dependencies ⇒ intercept those calls and returns dummy data.

+) Integration Tests: Kiểm thử tích hợp
- Kiểm tra xem các module hoạt động cùng nhau như thế nào. Verify "boundaries" and connection internal layers external
- How it work: Để các thành phần tương tác với nhau
- Ex: backend (NestJS): 1 intergration test gọi tới tầng service, truy cập PSQL ⇒ kiểm thử và xác minh

Scanned with CamScanner

+) End-to-End Tests: T
- Giả lập người dùng thực tế. UI → FE → BE → DB
- How it works: Đúng như users.

Tháp kiểm thử



Test Suite
Mock
Assertion
Test Runner
Flaky Test

- AI validation

+) The concept: Human might make syntax errox but, → AI violates logic

⇒ Solution: Use TDD to anchor AI ⇒ Write test first
⇒ AI have to make it pass the test

+) Testing as AI Guardrails
- Xác minh Logic nghiệp vụ (Business Logic): kiểm tra cùng các bất biến nghiệp vụ
- Phòng ngừa hồi quy (Guard Against Regression) → Tích hợp (Intergration) / Toàn trình (E+E)
- Khám phá TH ngoại lệ (Edge Case Discovery): Dùng AI viết test ngoại lệ LLMs

- CLI testing

- The core: It communicates with the terminal. End with Exit Code
- How to test:
  +) Sucess: Exit code: 0
  stdout: message
  +) Failed: _____ : non-0
  stdout: < stderr: message

write test first → Prompt AI → AI gen Code → Test → Fail / Pass

Guardrails
Logic drift
Hallucination
Business Invariant
Regression

+) Command Parsing and Validation: Users can pass arguments many ways ⇒ verify arguments
- Aliases: myCli --recursive / -r same result
- Missing Requirement.

+) System mocking and Side effect: CLI tools { read, write, delete } → Ig run on PC could be slow and important
- How to test: Virtual file:
  Dry Runs ⇒ --dry-run → xem trước hành động xảy ra mà k tạo file thay đổi

Exit Code	Stdout (Standard Output)	
Flag	Stderr	Dry-run

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Part II — TDD Practice

The screenshots document a test-driven development cycle: an initial test run could not resolve a module import, followed by successful unit/integration and end-to-end runs. The first result is a test setup/import error rather than a failed assertion.

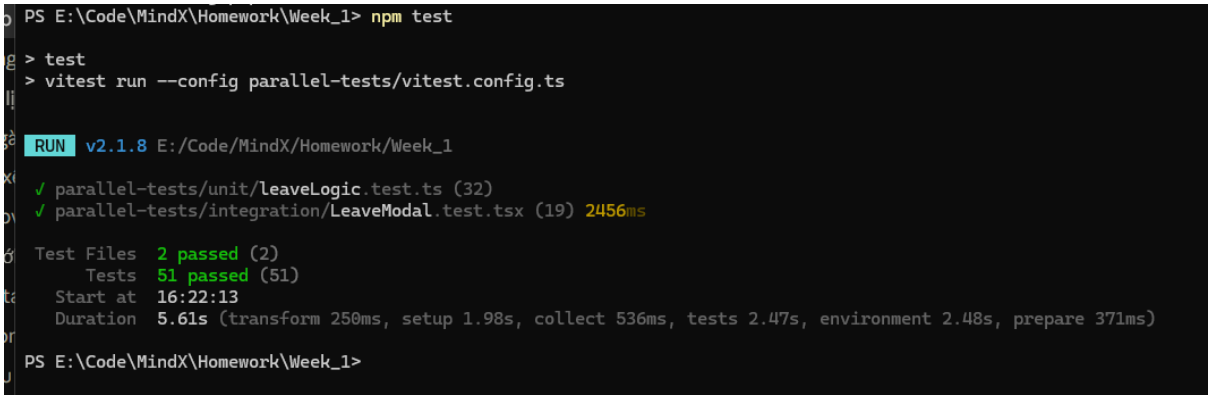
Red — Initial test run

The test runner cannot resolve the `leaveApi` module imported by the LeaveModal integration test.

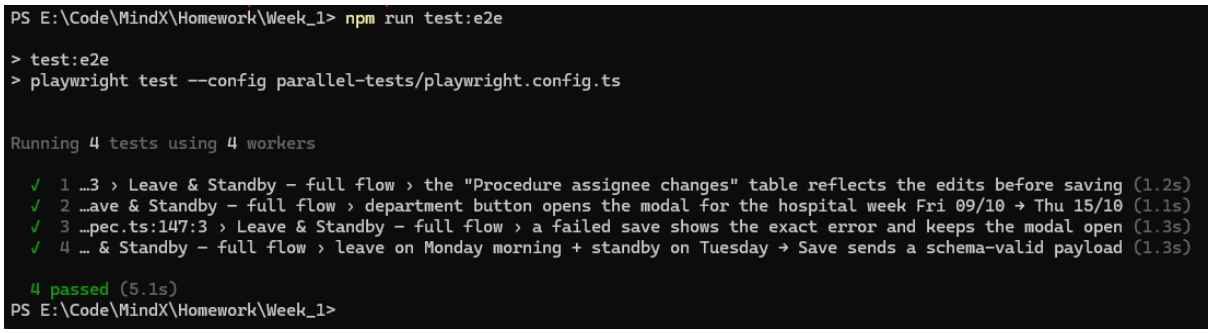


Green — Successful verification

The later unit/integration run reports 2 test files and 51 tests passed. The separate Playwright end-to-end run reports 4 tests passed.



Unit and integration tests: 51 passed across 2 files.



Playwright end-to-end tests: 4 passed.